4. Biological algicides Example: Bacteria
5. Destratification Example: physical/ mixing / forced aeration.

9.4 Water Pollution and Microbial Contamination

Water is polluted by both natural as well as man-made activities. Polluted water is one which consists of undesirable substances rendering it unfit for drinking and domestic use.

Sources of water pollutants

1. Industrial waste
2. Sewage waste
3. Mining activities
4. Marine dumping
5. Accidental oil leakage
6. Burning of fossil fuels
7. Chemical fertilizers and pesticides
8. Radio-active waste

The most prevalent biological contaminants in water are microbes, particularly bacteria and viruses. Most of the bacteria carried by storm runoff originate from animal fecal matter. Studies have shown that during storms, the water that drains off the land and into sewage systems also carries large quantity of bacteria and chemicals. The chemicals include pesticides applied to lawns, chemical wastes near industrial plants, and organic matter deposited o the ground by different sources. In addition to the chemical and biological contaminants, physical properties also affect the quality of biological life in water. Among these are pH, temperature, dissolved oxygen concentration and salinity.

Potable Water

Clean water free from odour, disagreeable taste, harmful chemicals, turbidity and microorganisms is called potable water, which is safe to drink and can use for food preparation without risk of health problems.

Biological oxygen demand (BOD)

Biological oxygen demand is one of the common parameter to monitor water quality and purity. BOD is the amount of dissolved oxygen needed by aerobic organisms to break down organic material present in a given water sample at certain temperature over a specific period of time. The amount of decomposable organic material in sewage is measured by the biochemical oxygen demand, or BOD. The BOD of the water sample is determined by aerating it, measuring the amount of oxygen in sample before incubation, placing the sample in a sealed container (BOD bottle) and incubating the container for five days at 20°C. During this five day period, microorganisms in the water grow and oxidize any organic materials in it. After incubation period, the BOD of the water can be determined by measuring the quantity of residual oxygen in the container. BOD of drinking water should be below 3ppm or 3mg/litre.

Indicator Microorganisms

Indicator organism are frequently used to monitor bacterial contamination of water. These indicator organisms provide a representative index of the water contamination by pathogenic microbes. The indicator organisms generally used in water quality monitoring are those that are associated with the gastrointestinal tract

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